Some backgrounds on D-FAST MI/BE discharge sets

1 Introduction

This memo gives some background on the set discharges to be used with D-FAST MI and with D-FAST BE. D-FAST MI is a tool for a rapid indication of morphological response to flood-plain interventions. It is appropriate as long as D-Morphology calculations are not necessary. In this memo, a set of discharges is proposed for application of D-FAST MI. Preferably, if both D-FAST MI and D-Morphology are applied, similar discharge situations should be considered. However, at the time of writing, the D-Morphology models are being developed and correspondence is not considered here.

D-FAST BE is a tool for indicating the erosion volumes of dynamic river banks. The discharges used are reviewed in this memo as well.

2 What is appropriate for D-FAST MI application?

D-FAST MI determines morphological effects with a fixed, pre-specified set of discharge situations. In principle, this set (via frequency and sequence of discharge situations) could be applied for scenarios with typically “wet,” “dry,” and “average” years. However, this option has not been verified yet and therefore will not be operational. Instead, the set of discharges is available in a preset, single annual hydrograph.

The discharge situations appropriate for D-FAST MI should reflect the hydraulic impact of an intervention in all flow regimes with mobile bed sediment. This implies defining discharges with a level of detail that matches the features of many potential interventions. However, a pragmatical selection is made, in analogy with the former Delft3D Rhine morphology models that used a set of eight discharges only.

Boven-Rijn discharge [m³/s] used in Delft3D simulations	probability of annual exceedance 1990-2020	equivalent set of Boven-Maas discharges [m³/s]
1203	0.887	38
1635	0.651	83
2250	0.326	225
3053	0.147	467
3824	0.079	663
4717	0.036	884
6151	0.012	1236
8592	0.001	2224

Tabel 1 Example of discharge-boundary set for former Delft3D Rijntakken model, with equivalent Meuse set.

3 Characteristic flow regimes

In the Rhine branches and Meuse, three typical regimes prevail due to the water level control by barriers;

1) barriers closed (apart from leakage and minor flow)
2) barriers gradually opening
3) barriers fully open

Additionally, the normalized river profiles enable distinguishing regimes with

i) main-channel flow only
ii) main-channel flow and developing floodplain flow
iii) main-channel flow and fully developed floodplain flow.

Since the barriers operate to control water levels in the main channel mainly, regimes 1) & 2) and i) coincide. This results in a classification represented in Table 2. Note that this is a simplification, since schedules of the barriers can differ.

regime	discharge Meuse [m³/s]	discharge Rhine [m³/s]
a) barriers closed, main-channel flow	<50	<1700
b) barriers opening, main-channel flow	50-1250	1700-3870
c) barriers fully open, main-channel flow and developing flow through the flood plains	1250-2300	3870-6000
d) barriers fully open, main-channel flow and developed flow through the flood plains	>2300	>6000

Table 2 Characterization of river-flow regimes.

In Table 2, four regimes are distinguished. The duration of each can be determined from historical or synthetical discharge series. Figure 1 illustrates the four regimes for Rhine and Meuse, using historical discharge series from 1990-2020.

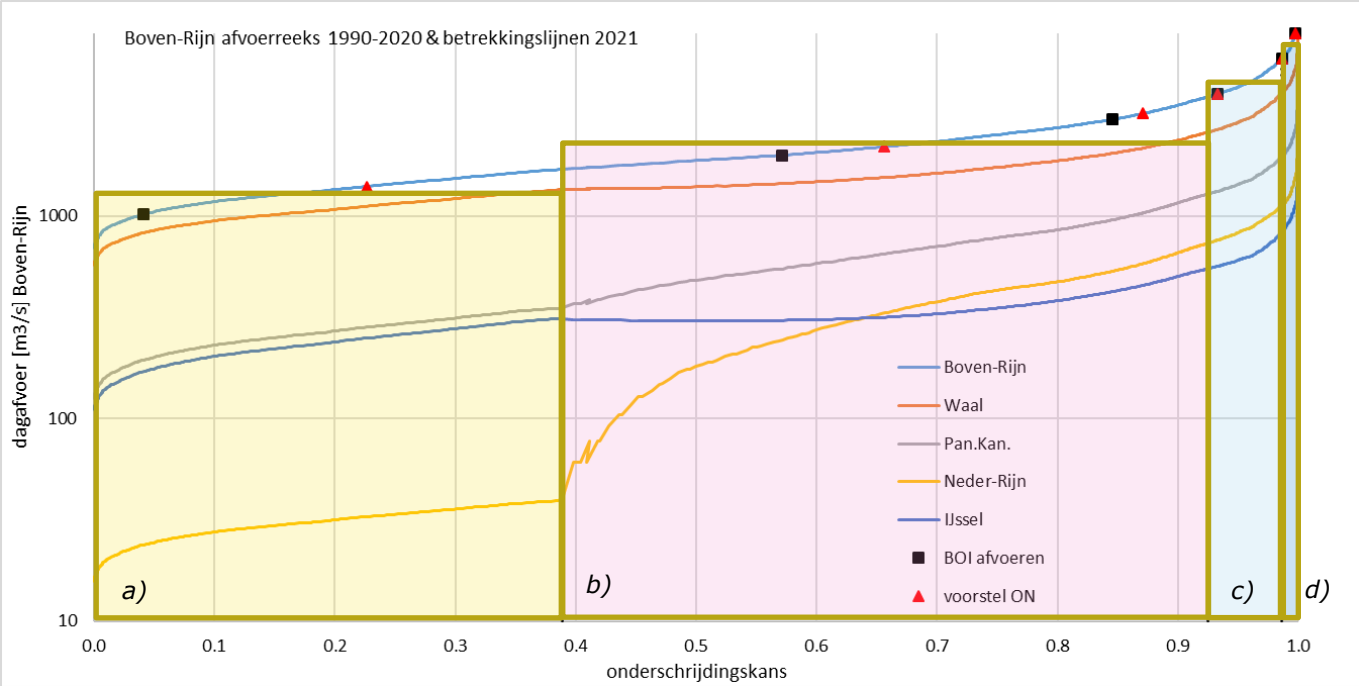

Figure 1-a Example of characteristic flow regimes Rhine.

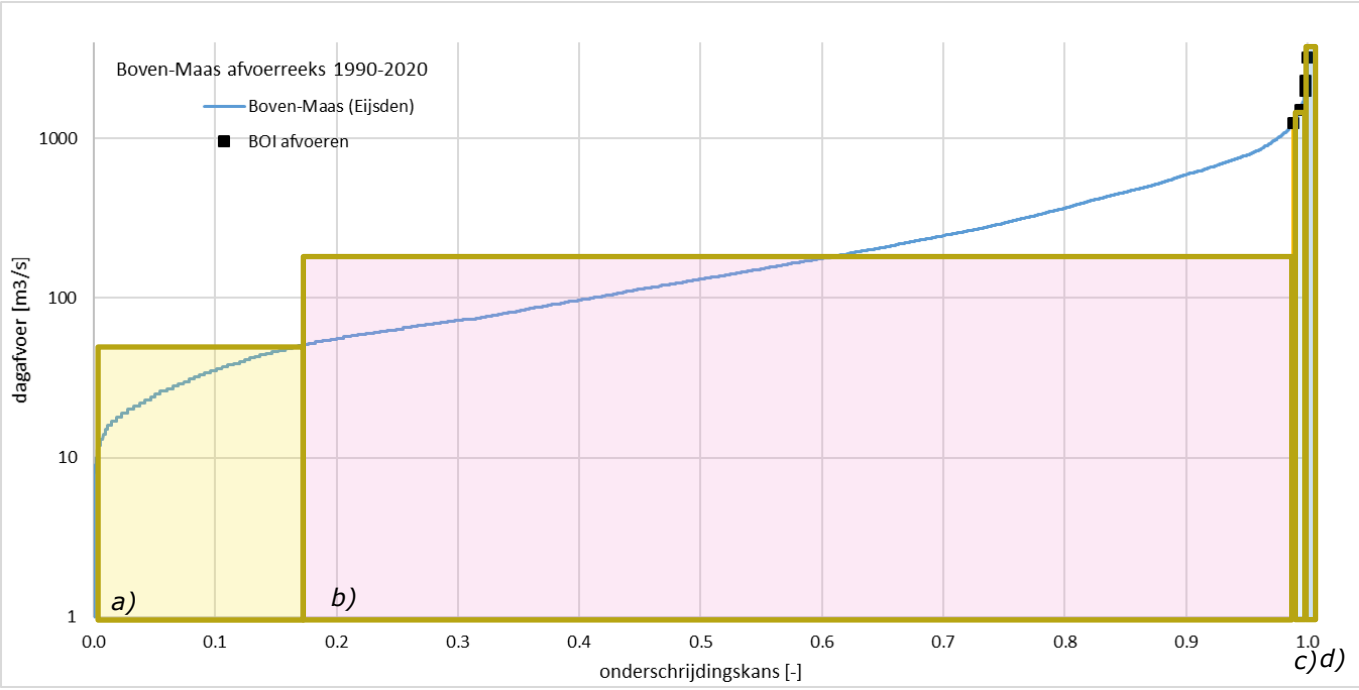

Figure 1-b Example of characteristic flow regimes Meuse.

4 Derivation of characteristic discharges from weir programs

D-FAST MI produces an estimate of the bed level response to intervention-induced changes in main-channel flow velocity. Since this hydraulic impact relates to the flow regime, every regime should be represented with at least one discharge, unless flow velocities are too small to induce sediment movement.

The morphological response is considered proportional to magnitude and duration of the intervention-induced forcing. Therefore, both magnitude and duration of a discharge is considered. This is illustrated as follows, with duration derived from a discharge exceedance curve $f(Q)$ [-] with $f(Q)$ the annual probability [-] of discharges exceeding a level Q [m^3/s].

<u>Regime a) barriers closed, main-channel flow</u>

The annual frequency of this regime is defined as $p_{barriers\ closed}$ [-], and characterized by for example a single discharge situation Q_1 [m^3/s] with frequency $p_{barriers\ closed}$ and $f(Q_1)$=1-$p_{barriers\ closed}$/2. In the period 1990-2020, this occurred for discharges below 1700 m^3/s in the Rhine ($p_{barriers\ closed} = 1 - f(1700) \approx 0.388$) and 50 m^3/s in the Meuse ($p_{barriers\ closed} = 1 - f(50) \approx 0.181$). This would yield a discharge Q_1=1336 m^3/s ($f(Q_1) \approx 0.806$) for the Rhine and Q_1=24 m^3/s ($f(Q_1) \approx 0.909$) for the Meuse.

<u>Regime b) barriers opening, main-channel flow</u>

In this regime, barriers change from closed to open. The frequency of barriers completely open is defined as $p_{barriers\ open}$ [-]. Hence, the frequency of the transition is 1- ($p_{barriers\ closed}$ + $p_{barriers\ open}$). Since hydraulic conditions are changing, it is appropriate to use at least two discharges for characterization. For example

Q_2 [m^3/s] with $f(Q_2)$= 1-$p_{barriers\ closed}$ - [1- $p_{barriers\ closed}$ - $p_{barriers\ open}$)]/3

Q_3 [m^3/s] with $f(Q_3)$= 1-$p_{barriers\ closed}$ - 2 [1- $p_{barriers\ closed}$ - $p_{barriers\ open}$)]/3

with frequencies [1- $p_{barriers\ closed}$ - $p_{barriers\ open}$]/2. For example, in 1990-2020 $p_{barriers\ open} = f(3870) \approx 0.076$ for the Rhine and $p_{barriers\ open} = f(1250) \approx 0.012$ for the Meuse. This would imply a frequency (1- $p_{barriers\ closed}$ - $p_{barriers\ open}$) equal to 0.536 for the Rhine and 0.807 for Meuse. This is illustrated in Table 3.

	$p_{barriers\ closed}$ [-]	$p_{barriers\ open}$ [-]	$f(Q_2)$ [-]	$f(Q_3)$ [-]	*frequency* [-]	Q_2 [m^3/s]	Q_3 [m^3/s]
Rhine	0.388	0.076	0.433	0.255	0.268	2000	2500
Meuse	0.181	0.012	0.555	0.281	0.403	150	300

Table 3 Example of representative discharges in regime b.

<u>c) barriers fully open, main channel flow and developing flow through the flood plains</u>

Since flow through flood plains develops with discharge rate, a frequency $p_{full\ convection}$ [-] is defined for fully developed flood plain convection. It is stated pragmatically that this prevails for discharges larger than 6000 m^3/s in the Rhine and 2300 m^3/s in the Meuse. In 1990-2020 this corresponds to $p_{full\ convection}$=f(6000 m^3/s) ≈0.014 for the Rhine and $p_{full\ convection}$=f(2300 m^3/s) ≈0.0003 for the Meuse. Hence, the frequency of this regime ($p_{weirs\ open} - p_{full\ convection}$) would be 0.062 for the Rhine and 0.0117 for the Meuse. Since frequencies are small, a single discharge is considered sufficient to represent this regime.

	$p_{barriers\ open}$ [-]	$p_{full\ convection}$ [-]	$f(Q_4)$ [-]	frequency [-]	Q_4 [m^3/s]
Rhine	0.076	0.014	0.045	0.062	4500
Meuse	0.012	0.0003	0.0062	0.012	1500

Table 4a Example of representative discharge in regime c.

<u>c) barriers fully open, main channel flow and developed flow through the flood plains</u>

In this regime of upper stages, flow convection through the flood plains is well developed.

A representative discharge could be defined as $f(Q_5)= p_{full\ convection}/2$. Since frequencies are small, a single discharge is considered sufficient to represent this regime.

	discharge level of developed convection	$p_{full\ convection}$ **[-]**	$f(Q_5)$ **[-]**	***frequency*** **[-]**	Q_5 **[m³/s]**
Rhine	6000	0.014	0.014	0.014	6725
Meuse	2300	0.0003	0.0003	0.0003	2600

Table 4b Example of representative discharge in regime d.

The resulting values of representative discharge are summarised in Table 5 and shown in Fig.2 for the Rhine and Fig.3 for the Meuse.

regime	**Rhine**	
	discharge [m³/s]	**frequency [-]**
a) barriers closed, main-channel flow	1336	0.388
b) barriers opening, main-channel flow	1900 -2700	0.269
c) barriers fully open, main-channel flow and developing flow through the flood plains	4473	0.061
d) barriers fully open, main-channel flow and developed flow through the flood plains	6724	0.014

regime	**Meuse**	
	discharge [m³/s]	**frequency [-]**
a) barriers closed, main-channel flow	32	0.086
b) barriers opening, main-channel flow	90 - 340	0.408
c) barriers fully open, main-channel flow and developing flow through the flood plains	1444	0.011
d) barriers fully open, main-channel flow and developed flow through the flood plains	2623	0.001

Table 5 Example of representative discharges based on weir programs.

5 Pragmatical selection of D-FAST MI discharges

Based on weir programs only, flow regimes can be characterized as represented in Table 5. However, for an adequate D-FAST MI application, the mobility of bed sediment should be considered as well, and from a practical point of view, it is efficient to aim for adopting BOI-discharges that are already used to evaluate flood safety as much as possible. The BOI-rates considered are 2000, 3000, 4000, 6000, 8000 m^3/s for the Rhine and 1300, 1700, 2100, 2500, 2800 and 3200 m^3/s for the Meuse.

Therefore, Figure 2 combines Table 5 with rates used by BOI and empirical values proposed by RWS Oost-Nederland. For high flow, the correspondence is clear, for low flows, the BOI rate is not representative for the entire regime. In branches with free flow this would affect the outcome. Hence, as far as the Rhine is considered, the BOI discharge set, including the lowest value proposed bij RWS Oost-Nederland, corresponds sufficiently with the characterization aimed for in Table 5. This yields a set consisting of 1300; 2000; 3000; 4000, 6000, and 8000 m^3/s.

Fig.3 refers to the Meuse. It shows that regimes c) and d) can be characterized by BOI discharge rates. Flood waves with short period exhibit dampening in the trajects of the Meuse. This would imply traject-specific reduced duration of peak discharges and elongation of less extreme discharges. That could affect D-FAST MI predictions in the lower Meuse reaches to a certain extend, but this effect is not considered here.

For low flows, no BOI rates are available. The mobility of river-bed sediment develops differently in weir-controlled and free-flowing reaches of the Meuse. In confined parts of reach km 16-68, computed flow velocities indicate mobility of sand volumes in the main channel at discharges above 50 m^3/s. At 125 m^3/s, sand and fine gravels become mobile in the entire reach, at 250 m^3/s medium coarse gravels start moving. At 500 m^3/s this applies for coarse gravel. In weir-controlled reaches downstream of km 68, sands become mobile at discharges above 200 m^3/s.

Hence, for Meuse traject km 16-68, discharges 50, 250, 500, 750, 1300, 2100, 2500 and 3200 m^3/s are considered representative for evaluating the dynamics of the relevant river-bed sediment. For Meuse trajects downstream of km 68, discharges 250, 500, 750, 1300, 2100, 2500 and 3200 m^3/s are considered representative.

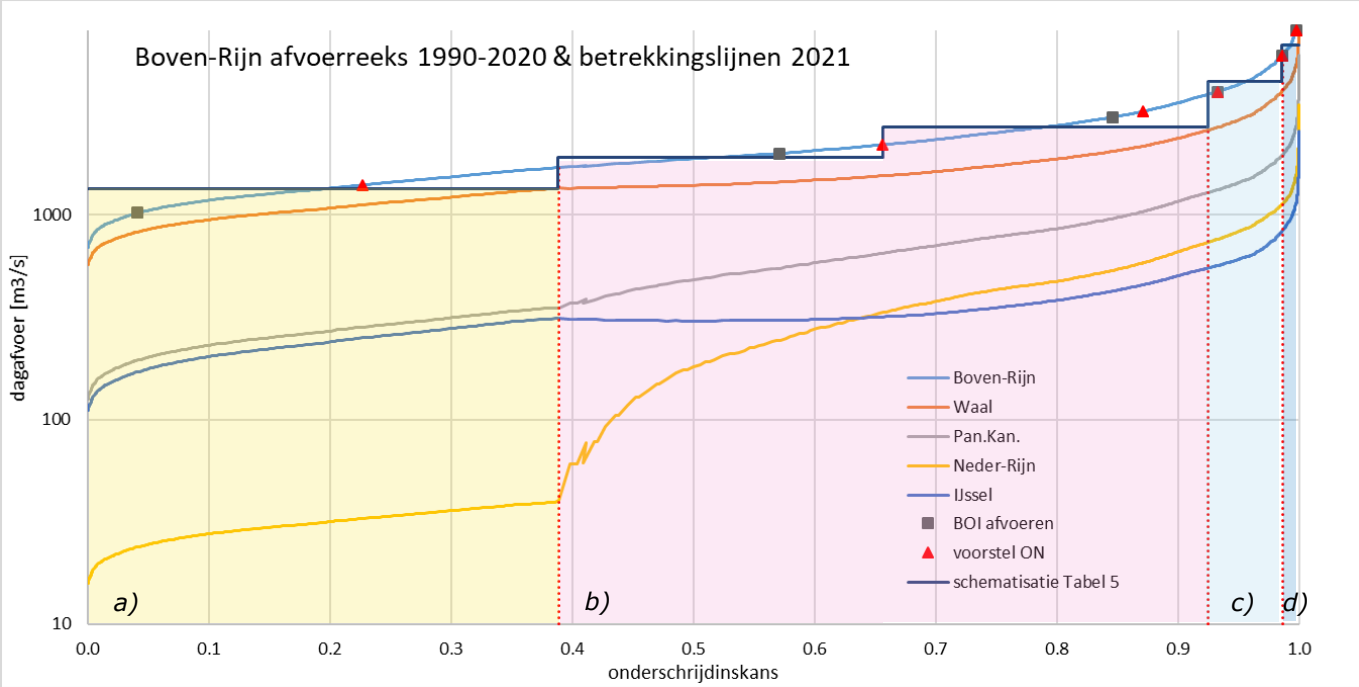

Figure 2 Discharge levels per flow regime Rhine.

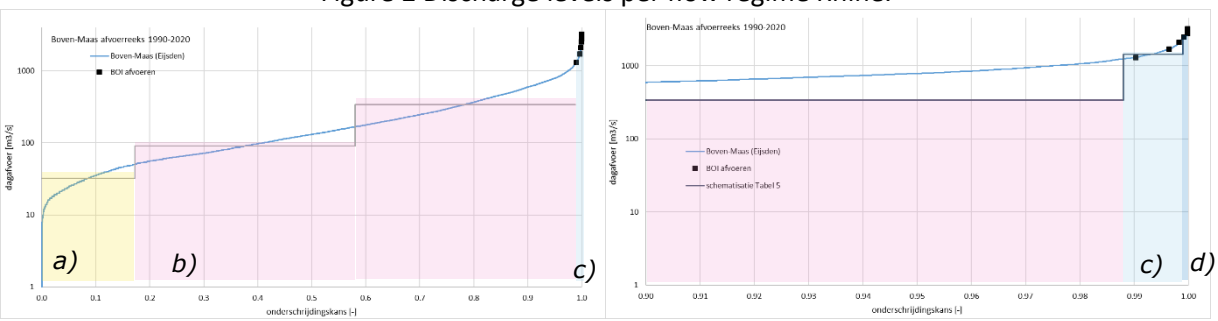

Figure 3 Discharge levels for the Meuse.

6 Conclusions D-FAST MI

Four typical flow regimes are distinguished. Characterizing these regimes and the expexted features if potential interventions, six discharge levels in the Rhine and six in the Meuse are suggested. For the Rhine it is proposed 1300; 2000; 3000; 4000, 6000, and 8000 m^3/s. Regarding Meuse traject km 16-68, the rate 50, 250, 500, 750, 1300, 2100, 2500 and 3200 m^3/s are considered representative. In trajects downstream of km 68, the set proposed is 250, 500, 750, 1300, 2100, 2500 and 3200 m^3/s.

7 Discharges in D-FAST BE

D-FAST BE is a tool for estimating annual erosion rates of dynamic river banks and equilibrium bank profiles. The hydraulic loads considered in D-FAST BE are stresses induced by ship-induced waves that dissipate on the banks and by the river flow over and along the bank. See D-FAST Bank Erosion User Manual (2022) for the relevant D-FAST BE discharges.

Dicharge level nr.	Discharge (m^3/s)	1998-2002 (wet)	2004-2010 (dry)	2008-2011 (intermediate)
1	75	0.2932	0.1808	0.3110
2	150	0.1918	0.2466	0.2329
3	272	0.1918	0.2603	0.2055
4	400	0.0411	0.0548	0.0685
5	500	0.1507	0.1370	0.0959
6	750	0.0548	0.0712	0.0548
7	900	0.0329	0.0384	0.0164
8	1100	0.0164	0.0082	0.0055
9	1300	0.0137	0.0014	0.0021
10	1500	0.0041	0.0014	0.0021
11	1700	0.0041	0.0000	0.0021
12	1900	0.0041	0.0000	0.0021
13	2300	0.0014	0.0000	0.0014

Table 3.1: Probabilities of a discharge level for different scenarios (De Vries, 2012)

Dicharge level nr.	Discharge (m^3/s)	average	16% undershot (wet)	50% undershot (intermediate)	84% undershot (dry)
1	1020	0.062	0.000	0.022	0.135
2	1203	0.160	0.066	0.193	0.245
3	1635	0.115	0.101	0.121	0.151
4	2250	0.200	0.154	0.238	0.189
5	3053	0.253	0.326	0.258	0.170
6	3824	0.116	0.179	0.086	0.063
7	4717	0.058	0.094	0.046	0.042
8	6151	0.027	0.061	0.026	0.006
9	8592	0.010	0.020	0.010	0.000

Table 3.2: Probabilities of discharge level in the Boven-Rijn (1989-2016) for different scenarios

Table 6 Discharges used in D-FAST BE.

The relevance of the rates in Table 6 is high-lighted as follows.

- The height and position of the river bank is identified as the set of grid cells that contain a land-water transition (depth = 0 m) at a reference discharge. This is used for georeferencing the equilibrium bank profiles predicted by D-FAST BE.
- The wave load is represented with wave height *H* [m] that varies with ship features and water depth. The area of impact is defined between *H*/2 above and 2*H* below the river-bank line. The frequency of water levels in this range determines the duration of bank erosion by wave load. *This requires considering flow conditions in a range around reference discharge.*
- In addition, D-FAST BE considers bank erosion by flow shear, with an erosion rate proportional to magnitude and frequency of flow velocity. *This implies considering discharges above the reference value.*

The D-FAST BE manual recommends using discharges that discretize the hydrograph, including the reference discharge that marks the river bank. In recent years, the first value in the table was considered as reference value. A 50% percentile value is a pragmatical option for a reference discharge. For a random series of 1990-2020 discharges, this would be 1877 m^3/s for the Rhine and 131 m^3/s for the Meuse.

This would motivate a D-FAST BE discharge set of 1020; 1300; 2000; 3000; 4000, 6000 and 8000 m^3/s (with 2000 m^3/s as reference) for the Rhine and 80; 130; 300; 750; 1300; 1700; 2100 and 2500 m^3/s (with 130 m^3/s as reference) for the Meuse.

8 Morphological time-scale

D-FAST MI predictions depend on morphological timescales. To ensure any tool-consistency, these time scales should be chosen such that D-FAST MI sufficiently mimics the dynamics simulated by morphological models. Since these models are in development yet, a preliminar set is derived here, based on rates used for WAQMORF.

The morphological time scale T_{mQ} [year] at a discharge rage Q [m^3/s] is defined here pragmatically as

$$T_{mQ} = 2\frac{B_{mc}}{w_Q} \qquad (1)$$

with a main-channel width B_{mc} [m] and a discharge-related, bed- level propagation rate w_Q [km/year]. The annual propagation of a bed level change $\overline{w}$ [m/year], is defined as

$$\overline{w}T = T\sum p_Q w_Q \qquad (2)$$

with T [year] and a discharge frequency p_Q [-]. To a certain extend, an indication for $\overline{w}$ can be derived empirically from bed-level soundings (third columns of Table 7, Table 8). The subsequent allocation over discharge is considered as follows.

For simplicity, propagation rates are approximated by bed-level celerities by
$w = \frac{u}{h}\frac{ds}{du} = b\frac{s}{h}$ with sediment transport s [m²/s] approximated by $s = mu^b$ (3)
with water depth h [m]; flow velocity u [m/s] and coefficients m [m^{2-b}/s^{1-b}] and b [-].
Eq.3 implies $w_Q = bm\frac{u_Q^b}{h_Q}$; $\overline{w} = \sum m\, b\, p_Q \frac{u_Q^b}{h_Q}$ and by combination $w_Q = \overline{w}\frac{u_Q^b}{h_Q}\left(\sum p_Q \frac{u_Q^b}{h_Q}\right)^{-1}$ (4)
The propagation rates w_Q are constructed in Table 7, with empirical indication of $\overline{w}$ (Table 8), an arbitrary b=5; frequencies p_Q that correspond to 1990-2020 discharge series and flow velocities u_Q [m/s] and water depths h_Q [m] generated by numerical river models.

reach	river km sections	$\overline{w}$ [km/year] 1974-1999	w_Q [km/year]					
			Q=1300 [m³/s]	Q=2000 [m³/s]	Q=3000 [m³/s]	Q=4000 [m³/s]	Q=6000 [m³/s]	Q=8000 [m³/s]
			p_Q=0.254 [-]	p_Q=0.563 [-]	p_Q=0.055 [-]	p_Q=0.106 [-]	p_Q=0.015 [-]	p_Q=0.007 [-]
Boven-Rijn	859-867	1.12	0.39	0.84	1.55	2.17	3.97	7.36
Boven-Waal	868-886	1.04	0.61	0.93	1.44	1.94	2.92	4.85
Midden-Waal	887-915	1.13	0.82	1.07	1.35	1.67	2.78	3.67
Beneden-Waal	916-951	1.10	0.49	0.93	1.61	2.41	4.34	5.82
Boven Merwede	951-961	1.08	0.07	0.35	1.44	4.23	12.80	20.79
Pannerdensch Kanaal	868-879	0.97	0.04	0.61	2.24	3.34	6.21	10.91
Neder-Rijn stuwpand Driel	880-891	0.77	0.00	0.35	3.43	4.71	4.99	6.06
Neder-Rijn stuwpand Amerongen	892-922	1.17	0.66	0.95	1.34	1.66	2.18	2.50
Lek stuwpand Hagestein	923-947	1.03	0.37	0.80	1.66	2.61	3.79	4.83
Lek *)	948-989	0.96	0.00	0.10	1.37	3.78	14.89	35.05
Boven-IJssel	880-930	1.50	1.07	1.49	1.79	2.08	3.00	3.52
Midden-IJssel	931-970	1.16	0.59	1.10	1.67	2.23	3.04	2.72
Beneden-IJssel	970-1000	1.06	0.20	0.68	1.59	3.27	7.46	11.24

reach	river km sections	$\overline{w}$ [km/jaar] 1950-1995	w_Q [km/year]								
			Q=50 [m³/s]	Q=250 [m³/s]	Q=500 [m³/s]	Q=750 [m³/s]	Q=1300 [m³/s]	Q=1700 [m³/s]	Q=2100 [m³/s]	Q=2500 [m³/s]	Q=3200 [m³/s]
			p_Q=0.3226	p_Q=0.3813	p_Q=0.1640	p_Q=0.0930	p_Q=0.0284	p_Q=0.0071	p_Q=0.0032	p_Q=0.0004	p_Q=0.0001
Grensmaas	16-69	0.001	0.0001	0.001	0.001	0.003	0.006	0.008	0.008	0.008	0.009

reach	river km sections	$\overline{w}$ [km/jaar] 1950-1995	w_Q [km/year]							
			Q=250 [m³/s]	Q=500 [m³/s]	Q=750 [m³/s]	Q=1300 [m³/s]	Q=1700 [m³/s]	Q=2100 [m³/s]	Q=2500 [m³/s]	Q=3200 [m³/s]
			p_Q=0.7716	p_Q=0.1245	p_Q=0.081	p_Q=0.0174	p_Q=0.0041	p_Q=0.0011	p_Q=0.0002	p_Q=0.0001
Linne-Roermond	69-80	0.14	0.015	0.206	0.780	1.796	1.602	1.268	0.421	0.159
Roermond-Belfeld	81-100	0.26	0.045	0.422	1.475	2.621	1.517	1.053	1.131	1.288
Belfeld-Sambeek	101-146	0.26	0.028	0.286	1.169	4.801	4.480	4.393	4.620	5.351
Sambeek-Grave	147-175	0.11	0.006	0.095	0.431	2.537	2.799	2.146	1.712	1.719
Grave-Lith	176-200	0.28	0.010	0.173	0.837	7.799	8.673	8.209	8.276	9.557
Lith-Ammerzoden	201-225	0.14	0.016	0.176	0.609	2.397	2.756	2.126	2.407	2.652
Ammerzoden-Keizersveer**)	225-247	0.14*	0.005	0.091	0.445	3.266	5.431	6.114	7.686	11.067

Table 7 Propagation rates proposed.

*) without tidal impact
**) no empirical indication available, assumed equal to traject Lith-Ammerzoden

flow velocities and depth derived from outdated river models
flow velocities and depth derived from recent D-Flow FM Rhine & Meuse models

reach	river km sections	$\bar{w}$ [km/year]		reference
		1950-1973	1974-1999	
Bovenrijn	859-867	0.97	1.12	RIZA werkdocument 2005.044x, Morfologische effecten Ruimte voor de Rivier in het Bovenrivierengebied
Boven-Waal	868-886	0.98	1.04	
Midden-Waal	887-915	1.09	1.13	
Beneden-Waal	916-951	1.15	1.10	
Pannerdensch Kanaal	868-879	0.79	0.97	
Nederrijn, stuwpand Driel	880-891	0.97	0.77	
Nederrijn, stuwpand Amerongen	892-922	1.36	1.17	
Lek, stuwpand Hagestein	923-947	1.16	1.03	
Lek	948-989	1.12	0.96	
Boven-IJssel	880-930	1.61	1.50	
Midden-IJssel	931-970	1.28	1.16	
Beneden-IJssel	970-1000	0.94	1.06	
		1975-2000		
Beneden Waal& Boven Merwede	934-961	1.08		RIZA WSR memo 2007-013 calibratie parameters Merwede.
Beneden Merwede	962-977	0.75		
Nieuwe Merwede	962-980	0.84		
		before 1950	after 1950	
Grensmaas (km 16-69)	16-69	0.001	0.001	Rijkswaterstaat Waterdienst report *Kennis en instrumenten Maas morfologie Inventarisatie behoefte monitoring en voorspelgereedschap)*
Linne-Roermond (km 69-80)	69-80	0.26	0.14	
Roermond-Belfeld (km 81-100)	81-100	0.23	0.26	
Belfeld Sambeek (km 101-146)	101-146	0.28	0.26	
Sambeek-Grave (km 147-175)	147-175	0.37	0.11	
Grave-Lith (km 176-200)	176-200	1.99	0.28	
Lith-Ammerzoden (km 201-227)	201-227	0.76	0.14	

Table 8 Indication year-averaged propagation rate (RWS memo *Methodiek inschatting morfologische effecten in het zomerbed door lokale rivieringrepen* *UPDATE December 2011*.)